

2650026_lab3_modified

September 18, 2026

Roll No.	Subject Code	Assignment No.
2650026	CS69101	Lab 3

To construct an AVL Tree for a given set of elements stored in a file, implement insertion and deletion operations on the constructed tree, and write the contents of the tree (after these operations) into a new file using in-order traversal.

0.1 Prerequisite Code Section

```
[1]: import sys
from io import TextIOWrapper
from typing import Self
import random
import sys
import pandas as pd
import matplotlib.pyplot as plt
from matplotlib.pylab import figure

sys.setrecursionlimit(100000) # Fix to avoid RecursionError

prep_arr = [random.randint(1, (i+1)*10) for i in range(5000)] # Random Array of 5000 Elements
# prep_arr.sort() # Sorted Array for Worst Case Scenario
print(prepare_arr[:25])
print(len(prepare_arr))

# Write the array to a file
f = open("./2650026_lab3_data/input0.txt", "w")
for i in prepare_arr:
    f.write(f"{i}, ")
f.write("\n")
f.close()

del prepare_arr
del f
```

```
iterations = 0
PRINT_DEBUG = False
```

```
[9, 9, 29, 18, 1, 53, 59, 50, 17, 20, 27, 118, 119, 49, 113, 145, 26, 155, 37,
30, 92, 159, 126, 100, 240]
5000
```

1 Q1

Read a set of integer elements from an input file (`input.txt`).

```
[2]: # Read a set of integer elements from an input file (input.txt)
arr0 = []
data0 = open("./2650026_lab3_data/input.txt", 'r')

try:
    for line in data0.readlines():
        line_data = line.rstrip().rstrip(',').replace(' ', '').split(',') #using_
↳rstrip to remove the \n
        for data in line_data:
            arr0.append(int(data))
    data0.close()
except Exception as e:
    sys.exit("[ERROR] Invalid data" + str(e))
```

```
[3]: print(arr0)
```

```
[70, 50, 90, 77, 55, 348, 76, 67, 69, 420, 25, 911, 210, 108, 768, 1080, 720]
```

2 Q2 & Q3

Construct a balanced AVL tree by inserting these elements one by one, performing the necessary LL, RR, LR, RL rotations to maintain balance after every insertion.

Implement an `insert(key)` operation that adds a new node to the AVL tree while preserving the AVL balance property

(balance factor $\in \{-1, 0, 1\}$ for every node).

```
[4]: class AVL_Node: # AVL Node class
    def __init__(self):
        self.__lchild = None
        self.__rchild = None
        self.__value = None
        self.__parent = None
        self.__height = None
        self.__balance_factor = None
```

```

def update_height_and_balance(self) -> None:
    # Print statement for debugging
    if PRINT_DEBUG:
        print(f"Updating height and balance factor for node with value: {self.__value}")
        print(f"Left child: {self.__lchild}, Right child: {self.__rchild}")
        self.__height = 1 + max(
            int(-1 if self.__lchild is None else self.__lchild.__get_height()),
            int(-1 if self.__rchild is None else self.__rchild.__get_height())
        )
        if PRINT_DEBUG:
            print(f"Updated height: {self.__height}")
        self.__balance_factor = (
            (-1 if self.__lchild is None else int(self.__lchild.__get_height())) -
            (-1 if self.__rchild is None else int(self.__rchild.__get_height()))
        ) # type: ignore
        if PRINT_DEBUG:
            print(f"Updated balance factor: {self.__balance_factor}")
        return

def __get_height(self) -> int:
    # temp_lchild = self.__lchild
    # temp_rchild = self.__rchild
    # temp_lchild_height = -1
    # temp_rchild_height = -1
    # if temp_lchild is not None:
    #     temp_lchild_height = temp_lchild.__get_height()
    # if temp_rchild is not None:
    #     temp_rchild_height = temp_rchild.__get_height()
    # # if self.__height is None:
    # self.__height = 1 + max(temp_lchild_height, temp_rchild_height)

    # Update the height if None
    if self.__height is None:
        self.update_height_and_balance()
        # print(f"Node value: {self.__value}, Left child height: {temp_lchild_height}, Right child height: {temp_rchild_height}, Node height: {self.__height}")
        return 0 if self.__height is None else self.__height

def __set_height(self, height: int) -> None:
    self.__height = height

height = property(__get_height, __set_height, doc="Height of the node")

```

```

def __get_balance_factor(self) -> int:
    return 0 if self.__balance_factor is None else self.__balance_factor

def __set_balance_factor(self, balance_factor: int) -> None:
    self.__balance_factor = balance_factor

balance_factor = property(__get_balance_factor, __set_balance_factor,
↪doc="Balance factor of the node")

def __get_lchild(self) -> Self | None:
    return self.__lchild

def __set_lchild(self, lchild: Self | None) -> None: # type: ignore
    if PRINT_DEBUG:
        print(f"Setting left child of node with value {self.__value} to_
↪{lchild}")
        self.__lchild = lchild
        if lchild is not None:
            lchild.__parent = self
            lchild.update_height_and_balance() # Update the height and balance_
↪factor of the left child when it is set
        self.update_height_and_balance() # Update the height and balance_
↪factor of the current node when the left child is set

lchild = property(__get_lchild, __set_lchild, doc="Left child of the node")

def __get_rchild(self) -> Self | None:
    return self.__rchild

def __set_rchild(self, rchild: Self | None) -> None: # type: ignore
    if PRINT_DEBUG:
        print(f"Setting right child of node with value {self.__value} to_
↪{rchild}")
        self.__rchild = rchild
        if rchild is not None:
            rchild.__parent = self
            rchild.update_height_and_balance() # Update the height and balance_
↪factor of the right child when it is set
        self.update_height_and_balance() # Update the height and balance_
↪factor of the current node when the right child is set

rchild = property(__get_rchild, __set_rchild, doc="Right child of the node")

def __get_parent(self) -> Self | None:
    return self.__parent

```

```

def __set_parent(self, parent: Self | None) -> None: # type: ignore
    if PRINT_DEBUG:
        print(f"Setting parent of node with value {self.__value} to {parent}")
    self.__parent = parent
    if parent is not None:
        parent.update_height_and_balance() # Update the height and balance factor of the parent when it is set

parent = property(__get_parent, __set_parent, doc="Parent of the node")

def __get_value(self) -> int | None:
    return self.__value

def __set_value(self, value: int) -> None:
    self.__value = value

value = property(__get_value, __set_value, doc="Value of the node")

def __str__(self) -> str:
    return f"Node(value={self.__value}, height={self.__height}, balance_factor={self.__balance_factor})"

def __repr__(self) -> str:
    return self.__str__()

def AVL_right_rotate(k2_node: AVL_Node | None, root_node: AVL_Node | None) -> AVL_Node | None:
    if PRINT_DEBUG:
        print(f"Right Rotate: k2_node={k2_node}, root_node={root_node}")
    k1_node = k2_node.lchild if k2_node is not None else None
    if k1_node is None or k2_node is None:
        return root_node

    if PRINT_DEBUG:
        print(f"LL Case: k1_node={k1_node}, k2_node={k2_node}")

    # From Weiss' Data Structures and Algorithm Analysis in C++ (Page 156)
    # Combined CLRS Logic with Weiss' Logic to avoid bugs

    # SAFETY CODE
    TEMP_NODE = k1_node.rchild
    OG_PARENT = k2_node.parent
    WAS_RCHILD = bool(OG_PARENT is not None and OG_PARENT.rchild is k2_node)

    # CLRS: x.lchild = y.rchild

```

```

# CLRS: y.rchild = x
k1_node.rchild = k2_node
k2_node.lchild = TEMP_NODE

# Update Parents

# CLRS: if y.rchild is not None:
# CLRS:     y.rchild.parent = y
# CLRS: // However, y.rchild is now x.lchild, so we need to update x.
↪lchild's parent to y
if PRINT_DEBUG:
    print(f"Updating parent of {k2_node.lchild} to {k2_node}")
if k2_node.lchild is not None:
    k2_node.lchild.parent = k2_node

# CLRS: x.parent = y.parent
k1_node.parent = OG_PARENT

# CLRS: if y.parent is None:
# CLRS:     root_node = y
if OG_PARENT is None:
    # root_node = k1_node
    pass # The input node will always be the root_node for the called_
↪function.
# CLRS: elif y is y.parent.rchild:
# CLRS:     y.parent.rchild = x
elif WAS_RCHILD:
    OG_PARENT.rchild = k1_node
    OG_PARENT.update_height_and_balance() # Update height and balance_
↪factor of OG_PARENT
# CLRS: else:
# CLRS:     y.parent.lchild = x
else:
    OG_PARENT.lchild = k1_node
    OG_PARENT.update_height_and_balance() # Update height and balance_
↪factor of OG_PARENT

root_node = k1_node # The input node will always be the root_node for the_
↪called function.

# CLRS: y.parent = x
# k2_node.parent = k1_node (THIS IS AUTO HANDLED BY SETTER METHOD)

# if k1_node.rchild is not None:
#     k1_node.rchild.parent = k1_node

```

```

    # root_node = k1_node if OG_PARENT is None else root_node // Copilot told
    ↪ me this

    k1_node.update_height_and_balance() # Update height and balance factor of
    ↪ k1_node
    k2_node.update_height_and_balance() # Update height and balance factor of
    ↪ k2_node

    if k1_node.parent is not None:
        k1_node.parent.update_height_and_balance() # Update height and balance
        ↪ factor of k1_node's parent

    return root_node

def AVL_left_rotate(k1_node: AVL_Node | None, root_node: AVL_Node | None) ->
    ↪ AVL_Node | None:
    if PRINT_DEBUG:
        print(f"Left Rotate: k1_node={k1_node}, root_node={root_node}")
    k2_node = k1_node.rchild if k1_node is not None else None
    if k1_node is None or k2_node is None:
        return root_node

    if PRINT_DEBUG:
        print(f"RR Case: k1_node={k1_node}, k2_node={k2_node}")

    # From Weiss' Data Structures and Algorithm Analysis in C++ (Page 156)
    # Combined CLRS Logic with Weiss' Logic to avoid bugs

    # SAFETY CODE
    TEMP_NODE = k2_node.lchild
    OG_PARENT = k1_node.parent
    WAS_RCHILD = bool(OG_PARENT is not None and OG_PARENT.rchild is k1_node)

    # CLRS: x.rchild = y.lchild
    # CLRS: y.lchild = x
    k2_node.lchild = k1_node
    k1_node.rchild = TEMP_NODE

    # Update Parents

    # CLRS: if y.lchild is not None:
    # CLRS:     y.lchild.parent = x
    # CLRS: // However, y.lchild is now x.rchild, so we need to update x.
    ↪ rchild's parent to y
    if k1_node.rchild is not None:
        k1_node.rchild.parent = k1_node

```

```

# CLRS: y.parent = x.parent
k2_node.parent = OG_PARENT

# CLRS: if x.parent is None:
# CLRS:     root_node = y
if OG_PARENT is None:
    # root_node = k2_node
    pass # The input node will always be the root_node for the called
↪function.
# CLRS: elseif x is x.parent.lchild:
# CLRS:     x.parent.lchild = y
elif WAS_RCHILD:
    OG_PARENT.rchild = k2_node
    OG_PARENT.update_height_and_balance() # Update height and balance
↪factor of OG_PARENT
# CLRS: else:
# CLRS:     x.parent.rchild = y
else:
    OG_PARENT.lchild = k2_node
    OG_PARENT.update_height_and_balance() # Update height and balance
↪factor of OG_PARENT

root_node = k2_node

# CLRS: x.parent = y
# k1_node.parent = k2_node (THIS IS AUTO HANDLED BY SETTER METHOD)

k1_node.update_height_and_balance() # Update height and balance factor of
↪k1_node
k2_node.update_height_and_balance() # Update height and balance factor of
↪k2_node

if k2_node.parent is not None:
    k2_node.parent.update_height_and_balance() # Update height and balance
↪factor of k2_node's parent

return root_node

#   k3
#   /
# k1      =>      k2
#  \            / \
#   k2         k1 k3

```

```

def AVL_double_rotate_right(k3_node: AVL_Node | None, root_node: AVL_Node |
↳None) -> AVL_Node | None:
    if PRINT_DEBUG:
        print(f"Double Right Rotate: k3_node={k3_node}, root_node={root_node}")
    k1_node = k3_node.lchild if k3_node is not None else None
    if k1_node is None or k3_node is None:
        return root_node

    if PRINT_DEBUG:
        print(f"Double Right Rotate: k1_node={k1_node}, k3_node={k3_node}")

    root_node = AVL_left_rotate(k1_node, root_node)
    root_node = AVL_right_rotate(k3_node, root_node)

    k3_node.update_height_and_balance() # Update height and balance factor of
↳k3_node
    k1_node.update_height_and_balance() # Update height and balance factor of
↳k1_node

    if k3_node.parent is not None:
        k3_node.parent.update_height_and_balance() # Update height and balance
↳factor of k3_node's parent

    return root_node

# k3
#   \
#   k1      =>      k2
#   /          / \
#  k2         k1 k3

def AVL_double_rotate_left(k3_node: AVL_Node | None, root_node: AVL_Node |
↳None) -> AVL_Node | None:
    if PRINT_DEBUG:
        print(f"Double Left Rotate: k3_node={k3_node}, root_node={root_node}")
    k1_node = k3_node.rchild if k3_node is not None else None
    if k1_node is None or k3_node is None:
        return root_node

    if PRINT_DEBUG:
        print(f"Double Left Rotate: k1_node={k1_node}, k3_node={k3_node}")

    root_node = AVL_right_rotate(k1_node, root_node)
    root_node = AVL_left_rotate(k3_node, root_node)

    k3_node.update_height_and_balance() # Update height and balance factor of
↳k3_node

```

```

    k1_node.update_height_and_balance() # Update height and balance factor of
↪k1_node

    if k3_node.parent is not None:
        k3_node.parent.update_height_and_balance() # Update height and balance
↪factor of k3_node's parent

    return root_node

def AVL_insert(root: AVL_Node | None, value: int) -> AVL_Node | None:
    if (root is None) or (root.value is None):
↪        # NULL type handling && Base Case
        new_node = AVL_Node()
        new_node.value = value
        # new_node.update_height_and_balance()
        if PRINT_DEBUG:
            print(f"Created {new_node}")
        return new_node

    if value < root.value:
↪        # X < MID
        root.lchild = AVL_insert(root.lchild, value)
        if PRINT_DEBUG:
            print(f"Inserted {root.lchild} as left child of {root}")
        if root.balance_factor == 2:
            if value < root.lchild.value: # type: ignore
                root = AVL_right_rotate(root, root)
                root.update_height_and_balance() if root is not None else None
↪# Update height and balance factor of root after rotation
            else:
                root = AVL_double_rotate_right(root, root)
                root.update_height_and_balance() if root is not None else None
↪# Update height and balance factor of root after rotation
        elif value > root.value:
↪        # X > MID
        root.rchild = AVL_insert(root.rchild, value)
        if PRINT_DEBUG:
            print(f"Inserted {root.rchild} as right child of {root}")
        if root.balance_factor == -2:
            if value > root.rchild.value: # type: ignore
                root = AVL_left_rotate(root, root)
                root.update_height_and_balance() if root is not None else None
↪# Update height and balance factor of root after rotation
            else:
                root = AVL_double_rotate_left(root, root)

```

```

        root.update_height_and_balance() if root is not None else None
    ↪# Update height and balance factor of root after rotation
    else:
        if PRINT_DEBUG:
            print(f"Value {value} already exists in the AVL tree. Duplicate
    ↪values are not allowed.")
        else:
            pass
    return root

```

```

[5]: PRINT_DEBUG = True
root0 = AVL_Node()
for i in arr0:
    root0 = AVL_insert(root0, i)
PRINT_DEBUG = False

```

```

Created Node(value=70, height=None, balance_factor=None)
Created Node(value=50, height=None, balance_factor=None)
Setting left child of node with value 70 to Node(value=50, height=None,
balance_factor=None)
Updating height and balance factor for node with value: 50
Left child: None, Right child: None
Updated height: 0
Updated balance factor: 0
Updating height and balance factor for node with value: 70
Left child: Node(value=50, height=0, balance_factor=0), Right child: None
Updated height: 1
Updated balance factor: 1
Inserted Node(value=50, height=0, balance_factor=0) as left child of
Node(value=70, height=1, balance_factor=1)
Created Node(value=90, height=None, balance_factor=None)
Setting right child of node with value 70 to Node(value=90, height=None,
balance_factor=None)
Updating height and balance factor for node with value: 90
Left child: None, Right child: None
Updated height: 0
Updated balance factor: 0
Updating height and balance factor for node with value: 70
Left child: Node(value=50, height=0, balance_factor=0), Right child:
Node(value=90, height=0, balance_factor=0)
Updated height: 1
Updated balance factor: 0
Inserted Node(value=90, height=0, balance_factor=0) as right child of
Node(value=70, height=1, balance_factor=0)
Created Node(value=77, height=None, balance_factor=None)
Setting left child of node with value 90 to Node(value=77, height=None,
balance_factor=None)
Updating height and balance factor for node with value: 77

```

Left child: None, Right child: None
 Updated height: 0
 Updated balance factor: 0
 Updating height and balance factor for node with value: 90
 Left child: Node(value=77, height=0, balance_factor=0), Right child: None
 Updated height: 1
 Updated balance factor: 1
 Inserted Node(value=77, height=0, balance_factor=0) as left child of
 Node(value=90, height=1, balance_factor=1)
 Setting right child of node with value 70 to Node(value=90, height=1,
 balance_factor=1)
 Updating height and balance factor for node with value: 90
 Left child: Node(value=77, height=0, balance_factor=0), Right child: None
 Updated height: 1
 Updated balance factor: 1
 Updating height and balance factor for node with value: 70
 Left child: Node(value=50, height=0, balance_factor=0), Right child:
 Node(value=90, height=1, balance_factor=1)
 Updated height: 2
 Updated balance factor: -1
 Inserted Node(value=90, height=1, balance_factor=1) as right child of
 Node(value=70, height=2, balance_factor=-1)
 Created Node(value=55, height=None, balance_factor=None)
 Setting right child of node with value 50 to Node(value=55, height=None,
 balance_factor=None)
 Updating height and balance factor for node with value: 55
 Left child: None, Right child: None
 Updated height: 0
 Updated balance factor: 0
 Updating height and balance factor for node with value: 50
 Left child: None, Right child: Node(value=55, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: -1
 Inserted Node(value=55, height=0, balance_factor=0) as right child of
 Node(value=50, height=1, balance_factor=-1)
 Setting left child of node with value 70 to Node(value=50, height=1,
 balance_factor=-1)
 Updating height and balance factor for node with value: 50
 Left child: None, Right child: Node(value=55, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: -1
 Updating height and balance factor for node with value: 70
 Left child: Node(value=50, height=1, balance_factor=-1), Right child:
 Node(value=90, height=1, balance_factor=1)
 Updated height: 2
 Updated balance factor: 0
 Inserted Node(value=50, height=1, balance_factor=-1) as left child of
 Node(value=70, height=2, balance_factor=0)

Created Node(value=348, height=None, balance_factor=None)
 Setting right child of node with value 90 to Node(value=348, height=None, balance_factor=None)
 Updating height and balance factor for node with value: 348
 Left child: None, Right child: None
 Updated height: 0
 Updated balance factor: 0
 Updating height and balance factor for node with value: 90
 Left child: Node(value=77, height=0, balance_factor=0), Right child: Node(value=348, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: 0
 Inserted Node(value=348, height=0, balance_factor=0) as right child of Node(value=90, height=1, balance_factor=0)
 Setting right child of node with value 70 to Node(value=90, height=1, balance_factor=0)
 Updating height and balance factor for node with value: 90
 Left child: Node(value=77, height=0, balance_factor=0), Right child: Node(value=348, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: 0
 Updating height and balance factor for node with value: 70
 Left child: Node(value=50, height=1, balance_factor=-1), Right child: Node(value=90, height=1, balance_factor=0)
 Updated height: 2
 Updated balance factor: 0
 Inserted Node(value=90, height=1, balance_factor=0) as right child of Node(value=70, height=2, balance_factor=0)
 Created Node(value=76, height=None, balance_factor=None)
 Setting left child of node with value 77 to Node(value=76, height=None, balance_factor=None)
 Updating height and balance factor for node with value: 76
 Left child: None, Right child: None
 Updated height: 0
 Updated balance factor: 0
 Updating height and balance factor for node with value: 77
 Left child: Node(value=76, height=0, balance_factor=0), Right child: None
 Updated height: 1
 Updated balance factor: 1
 Inserted Node(value=76, height=0, balance_factor=0) as left child of Node(value=77, height=1, balance_factor=1)
 Setting left child of node with value 90 to Node(value=77, height=1, balance_factor=1)
 Updating height and balance factor for node with value: 77
 Left child: Node(value=76, height=0, balance_factor=0), Right child: None
 Updated height: 1
 Updated balance factor: 1
 Updating height and balance factor for node with value: 90

Left child: Node(value=77, height=1, balance_factor=1), Right child:
 Node(value=348, height=0, balance_factor=0)
 Updated height: 2
 Updated balance factor: 1
 Inserted Node(value=77, height=1, balance_factor=1) as left child of
 Node(value=90, height=2, balance_factor=1)
 Setting right child of node with value 70 to Node(value=90, height=2,
 balance_factor=1)
 Updating height and balance factor for node with value: 90
 Left child: Node(value=77, height=1, balance_factor=1), Right child:
 Node(value=348, height=0, balance_factor=0)
 Updated height: 2
 Updated balance factor: 1
 Updating height and balance factor for node with value: 70
 Left child: Node(value=50, height=1, balance_factor=-1), Right child:
 Node(value=90, height=2, balance_factor=1)
 Updated height: 3
 Updated balance factor: -1
 Inserted Node(value=90, height=2, balance_factor=1) as right child of
 Node(value=70, height=3, balance_factor=-1)
 Created Node(value=67, height=None, balance_factor=None)
 Setting right child of node with value 55 to Node(value=67, height=None,
 balance_factor=None)
 Updating height and balance factor for node with value: 67
 Left child: None, Right child: None
 Updated height: 0
 Updated balance factor: 0
 Updating height and balance factor for node with value: 55
 Left child: None, Right child: Node(value=67, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: -1
 Inserted Node(value=67, height=0, balance_factor=0) as right child of
 Node(value=55, height=1, balance_factor=-1)
 Setting right child of node with value 50 to Node(value=55, height=1,
 balance_factor=-1)
 Updating height and balance factor for node with value: 55
 Left child: None, Right child: Node(value=67, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: -1
 Updating height and balance factor for node with value: 50
 Left child: None, Right child: Node(value=55, height=1, balance_factor=-1)
 Updated height: 2
 Updated balance factor: -2
 Inserted Node(value=55, height=1, balance_factor=-1) as right child of
 Node(value=50, height=2, balance_factor=-2)
 Left Rotate: k1_node=Node(value=50, height=2, balance_factor=-2),
 root_node=Node(value=50, height=2, balance_factor=-2)
 RR Case: k1_node=Node(value=50, height=2, balance_factor=-2),

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k2_node=Node(value=55, height=1, balance_factor=-1)
Setting left child of node with value 55 to Node(value=50, height=2,
balance_factor=-2)
Updating height and balance factor for node with value: 50
Left child: None, Right child: Node(value=55, height=1, balance_factor=-1)
Updated height: 2
Updated balance factor: -2
Updating height and balance factor for node with value: 55
Left child: Node(value=50, height=2, balance_factor=-2), Right child:
Node(value=67, height=0, balance_factor=0)
Updated height: 3
Updated balance factor: 2
Setting right child of node with value 50 to None
Updating height and balance factor for node with value: 50
Left child: None, Right child: None
Updated height: 0
Updated balance factor: 0
Setting parent of node with value 55 to Node(value=70, height=3,
balance_factor=-1)
Updating height and balance factor for node with value: 70
Left child: Node(value=50, height=0, balance_factor=0), Right child:
Node(value=90, height=2, balance_factor=1)
Updated height: 3
Updated balance factor: -2
Setting left child of node with value 70 to Node(value=55, height=3,
balance_factor=2)
Updating height and balance factor for node with value: 55
Left child: Node(value=50, height=0, balance_factor=0), Right child:
Node(value=67, height=0, balance_factor=0)
Updated height: 1
Updated balance factor: 0
Updating height and balance factor for node with value: 70
Left child: Node(value=55, height=1, balance_factor=0), Right child:
Node(value=90, height=2, balance_factor=1)
Updated height: 3
Updated balance factor: -1
Updating height and balance factor for node with value: 70
Left child: Node(value=55, height=1, balance_factor=0), Right child:
Node(value=90, height=2, balance_factor=1)
Updated height: 3
Updated balance factor: -1
Updating height and balance factor for node with value: 50
Left child: None, Right child: None
Updated height: 0
Updated balance factor: 0
Updating height and balance factor for node with value: 55
Left child: Node(value=50, height=0, balance_factor=0), Right child:
Node(value=67, height=0, balance_factor=0)

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Updated height: 1
 Updated balance factor: 0
 Updating height and balance factor for node with value: 70
 Left child: Node(value=55, height=1, balance_factor=0), Right child:
 Node(value=90, height=2, balance_factor=1)
 Updated height: 3
 Updated balance factor: -1
 Updating height and balance factor for node with value: 55
 Left child: Node(value=50, height=0, balance_factor=0), Right child:
 Node(value=67, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: 0
 Setting left child of node with value 70 to Node(value=55, height=1,
 balance_factor=0)
 Updating height and balance factor for node with value: 55
 Left child: Node(value=50, height=0, balance_factor=0), Right child:
 Node(value=67, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: 0
 Updating height and balance factor for node with value: 70
 Left child: Node(value=55, height=1, balance_factor=0), Right child:
 Node(value=90, height=2, balance_factor=1)
 Updated height: 3
 Updated balance factor: -1
 Inserted Node(value=55, height=1, balance_factor=0) as left child of
 Node(value=70, height=3, balance_factor=-1)
 Created Node(value=69, height=None, balance_factor=None)
 Setting right child of node with value 67 to Node(value=69, height=None,
 balance_factor=None)
 Updating height and balance factor for node with value: 69
 Left child: None, Right child: None
 Updated height: 0
 Updated balance factor: 0
 Updating height and balance factor for node with value: 67
 Left child: None, Right child: Node(value=69, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: -1
 Inserted Node(value=69, height=0, balance_factor=0) as right child of
 Node(value=67, height=1, balance_factor=-1)
 Setting right child of node with value 55 to Node(value=67, height=1,
 balance_factor=-1)
 Updating height and balance factor for node with value: 67
 Left child: None, Right child: Node(value=69, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: -1
 Updating height and balance factor for node with value: 55
 Left child: Node(value=50, height=0, balance_factor=0), Right child:
 Node(value=67, height=1, balance_factor=-1)

Updated height: 2
 Updated balance factor: -1
 Inserted Node(value=67, height=1, balance_factor=-1) as right child of
 Node(value=55, height=2, balance_factor=-1)
 Setting left child of node with value 70 to Node(value=55, height=2,
 balance_factor=-1)
 Updating height and balance factor for node with value: 55
 Left child: Node(value=50, height=0, balance_factor=0), Right child:
 Node(value=67, height=1, balance_factor=-1)
 Updated height: 2
 Updated balance factor: -1
 Updating height and balance factor for node with value: 70
 Left child: Node(value=55, height=2, balance_factor=-1), Right child:
 Node(value=90, height=2, balance_factor=1)
 Updated height: 3
 Updated balance factor: 0
 Inserted Node(value=55, height=2, balance_factor=-1) as left child of
 Node(value=70, height=3, balance_factor=0)
 Created Node(value=420, height=None, balance_factor=None)
 Setting right child of node with value 348 to Node(value=420, height=None,
 balance_factor=None)
 Updating height and balance factor for node with value: 420
 Left child: None, Right child: None
 Updated height: 0
 Updated balance factor: 0
 Updating height and balance factor for node with value: 348
 Left child: None, Right child: Node(value=420, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: -1
 Inserted Node(value=420, height=0, balance_factor=0) as right child of
 Node(value=348, height=1, balance_factor=-1)
 Setting right child of node with value 90 to Node(value=348, height=1,
 balance_factor=-1)
 Updating height and balance factor for node with value: 348
 Left child: None, Right child: Node(value=420, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: -1
 Updating height and balance factor for node with value: 90
 Left child: Node(value=77, height=1, balance_factor=1), Right child:
 Node(value=348, height=1, balance_factor=-1)
 Updated height: 2
 Updated balance factor: 0
 Inserted Node(value=348, height=1, balance_factor=-1) as right child of
 Node(value=90, height=2, balance_factor=0)
 Setting right child of node with value 70 to Node(value=90, height=2,
 balance_factor=0)
 Updating height and balance factor for node with value: 90
 Left child: Node(value=77, height=1, balance_factor=1), Right child:

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Node(value=348, height=1, balance_factor=-1)
Updated height: 2
Updated balance factor: 0
Updating height and balance factor for node with value: 70
Left child: Node(value=55, height=2, balance_factor=-1), Right child:
Node(value=90, height=2, balance_factor=0)
Updated height: 3
Updated balance factor: 0
Inserted Node(value=90, height=2, balance_factor=0) as right child of
Node(value=70, height=3, balance_factor=0)
Created Node(value=25, height=None, balance_factor=None)
Setting left child of node with value 50 to Node(value=25, height=None,
balance_factor=None)
Updating height and balance factor for node with value: 25
Left child: None, Right child: None
Updated height: 0
Updated balance factor: 0
Updating height and balance factor for node with value: 50
Left child: Node(value=25, height=0, balance_factor=0), Right child: None
Updated height: 1
Updated balance factor: 1
Inserted Node(value=25, height=0, balance_factor=0) as left child of
Node(value=50, height=1, balance_factor=1)
Setting left child of node with value 55 to Node(value=50, height=1,
balance_factor=1)
Updating height and balance factor for node with value: 50
Left child: Node(value=25, height=0, balance_factor=0), Right child: None
Updated height: 1
Updated balance factor: 1
Updating height and balance factor for node with value: 55
Left child: Node(value=50, height=1, balance_factor=1), Right child:
Node(value=67, height=1, balance_factor=-1)
Updated height: 2
Updated balance factor: 0
Inserted Node(value=50, height=1, balance_factor=1) as left child of
Node(value=55, height=2, balance_factor=0)
Setting left child of node with value 70 to Node(value=55, height=2,
balance_factor=0)
Updating height and balance factor for node with value: 55
Left child: Node(value=50, height=1, balance_factor=1), Right child:
Node(value=67, height=1, balance_factor=-1)
Updated height: 2
Updated balance factor: 0
Updating height and balance factor for node with value: 70
Left child: Node(value=55, height=2, balance_factor=0), Right child:
Node(value=90, height=2, balance_factor=0)
Updated height: 3
Updated balance factor: 0

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Inserted Node(value=55, height=2, balance_factor=0) as left child of
Node(value=70, height=3, balance_factor=0)
Created Node(value=911, height=None, balance_factor=None)
Setting right child of node with value 420 to Node(value=911, height=None,
balance_factor=None)
Updating height and balance factor for node with value: 911
Left child: None, Right child: None
Updated height: 0
Updated balance factor: 0
Updating height and balance factor for node with value: 420
Left child: None, Right child: Node(value=911, height=0, balance_factor=0)
Updated height: 1
Updated balance factor: -1
Inserted Node(value=911, height=0, balance_factor=0) as right child of
Node(value=420, height=1, balance_factor=-1)
Setting right child of node with value 348 to Node(value=420, height=1,
balance_factor=-1)
Updating height and balance factor for node with value: 420
Left child: None, Right child: Node(value=911, height=0, balance_factor=0)
Updated height: 1
Updated balance factor: -1
Updating height and balance factor for node with value: 348
Left child: None, Right child: Node(value=420, height=1, balance_factor=-1)
Updated height: 2
Updated balance factor: -2
Inserted Node(value=420, height=1, balance_factor=-1) as right child of
Node(value=348, height=2, balance_factor=-2)
Left Rotate: k1_node=Node(value=348, height=2, balance_factor=-2),
root_node=Node(value=348, height=2, balance_factor=-2)
RR Case: k1_node=Node(value=348, height=2, balance_factor=-2),
k2_node=Node(value=420, height=1, balance_factor=-1)
Setting left child of node with value 420 to Node(value=348, height=2,
balance_factor=-2)
Updating height and balance factor for node with value: 348
Left child: None, Right child: Node(value=420, height=1, balance_factor=-1)
Updated height: 2
Updated balance factor: -2
Updating height and balance factor for node with value: 420
Left child: Node(value=348, height=2, balance_factor=-2), Right child:
Node(value=911, height=0, balance_factor=0)
Updated height: 3
Updated balance factor: 2
Setting right child of node with value 348 to None
Updating height and balance factor for node with value: 348
Left child: None, Right child: None
Updated height: 0
Updated balance factor: 0
Setting parent of node with value 420 to Node(value=90, height=2,

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balance_factor=0)
Updating height and balance factor for node with value: 90
Left child: Node(value=77, height=1, balance_factor=1), Right child:
Node(value=348, height=0, balance_factor=0)
Updated height: 2
Updated balance factor: 1
Setting right child of node with value 90 to Node(value=420, height=3,
balance_factor=2)
Updating height and balance factor for node with value: 420
Left child: Node(value=348, height=0, balance_factor=0), Right child:
Node(value=911, height=0, balance_factor=0)
Updated height: 1
Updated balance factor: 0
Updating height and balance factor for node with value: 90
Left child: Node(value=77, height=1, balance_factor=1), Right child:
Node(value=420, height=1, balance_factor=0)
Updated height: 2
Updated balance factor: 0
Updating height and balance factor for node with value: 90
Left child: Node(value=77, height=1, balance_factor=1), Right child:
Node(value=420, height=1, balance_factor=0)
Updated height: 2
Updated balance factor: 0
Updating height and balance factor for node with value: 348
Left child: None, Right child: None
Updated height: 0
Updated balance factor: 0
Updating height and balance factor for node with value: 420
Left child: Node(value=348, height=0, balance_factor=0), Right child:
Node(value=911, height=0, balance_factor=0)
Updated height: 1
Updated balance factor: 0
Updating height and balance factor for node with value: 90
Left child: Node(value=77, height=1, balance_factor=1), Right child:
Node(value=420, height=1, balance_factor=0)
Updated height: 2
Updated balance factor: 0
Updating height and balance factor for node with value: 420
Left child: Node(value=348, height=0, balance_factor=0), Right child:
Node(value=911, height=0, balance_factor=0)
Updated height: 1
Updated balance factor: 0
Setting right child of node with value 90 to Node(value=420, height=1,
balance_factor=0)
Updating height and balance factor for node with value: 420
Left child: Node(value=348, height=0, balance_factor=0), Right child:
Node(value=911, height=0, balance_factor=0)
Updated height: 1

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Updated balance factor: 0
 Updating height and balance factor for node with value: 90
 Left child: Node(value=77, height=1, balance_factor=1), Right child:
 Node(value=420, height=1, balance_factor=0)
 Updated height: 2
 Updated balance factor: 0
 Inserted Node(value=420, height=1, balance_factor=0) as right child of
 Node(value=90, height=2, balance_factor=0)
 Setting right child of node with value 70 to Node(value=90, height=2,
 balance_factor=0)
 Updating height and balance factor for node with value: 90
 Left child: Node(value=77, height=1, balance_factor=1), Right child:
 Node(value=420, height=1, balance_factor=0)
 Updated height: 2
 Updated balance factor: 0
 Updating height and balance factor for node with value: 70
 Left child: Node(value=55, height=2, balance_factor=0), Right child:
 Node(value=90, height=2, balance_factor=0)
 Updated height: 3
 Updated balance factor: 0
 Inserted Node(value=90, height=2, balance_factor=0) as right child of
 Node(value=70, height=3, balance_factor=0)
 Created Node(value=210, height=None, balance_factor=None)
 Setting left child of node with value 348 to Node(value=210, height=None,
 balance_factor=None)
 Updating height and balance factor for node with value: 210
 Left child: None, Right child: None
 Updated height: 0
 Updated balance factor: 0
 Updating height and balance factor for node with value: 348
 Left child: Node(value=210, height=0, balance_factor=0), Right child: None
 Updated height: 1
 Updated balance factor: 1
 Inserted Node(value=210, height=0, balance_factor=0) as left child of
 Node(value=348, height=1, balance_factor=1)
 Setting left child of node with value 420 to Node(value=348, height=1,
 balance_factor=1)
 Updating height and balance factor for node with value: 348
 Left child: Node(value=210, height=0, balance_factor=0), Right child: None
 Updated height: 1
 Updated balance factor: 1
 Updating height and balance factor for node with value: 420
 Left child: Node(value=348, height=1, balance_factor=1), Right child:
 Node(value=911, height=0, balance_factor=0)
 Updated height: 2
 Updated balance factor: 1
 Inserted Node(value=348, height=1, balance_factor=1) as left child of
 Node(value=420, height=2, balance_factor=1)

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Setting right child of node with value 90 to Node(value=420, height=2,
balance_factor=1)
Updating height and balance factor for node with value: 420
Left child: Node(value=348, height=1, balance_factor=1), Right child:
Node(value=911, height=0, balance_factor=0)
Updated height: 2
Updated balance factor: 1
Updating height and balance factor for node with value: 90
Left child: Node(value=77, height=1, balance_factor=1), Right child:
Node(value=420, height=2, balance_factor=1)
Updated height: 3
Updated balance factor: -1
Inserted Node(value=420, height=2, balance_factor=1) as right child of
Node(value=90, height=3, balance_factor=-1)
Setting right child of node with value 70 to Node(value=90, height=3,
balance_factor=-1)
Updating height and balance factor for node with value: 90
Left child: Node(value=77, height=1, balance_factor=1), Right child:
Node(value=420, height=2, balance_factor=1)
Updated height: 3
Updated balance factor: -1
Updating height and balance factor for node with value: 70
Left child: Node(value=55, height=2, balance_factor=0), Right child:
Node(value=90, height=3, balance_factor=-1)
Updated height: 4
Updated balance factor: -1
Inserted Node(value=90, height=3, balance_factor=-1) as right child of
Node(value=70, height=4, balance_factor=-1)
Created Node(value=108, height=None, balance_factor=None)
Setting left child of node with value 210 to Node(value=108, height=None,
balance_factor=None)
Updating height and balance factor for node with value: 108
Left child: None, Right child: None
Updated height: 0
Updated balance factor: 0
Updating height and balance factor for node with value: 210
Left child: Node(value=108, height=0, balance_factor=0), Right child: None
Updated height: 1
Updated balance factor: 1
Inserted Node(value=108, height=0, balance_factor=0) as left child of
Node(value=210, height=1, balance_factor=1)
Setting left child of node with value 348 to Node(value=210, height=1,
balance_factor=1)
Updating height and balance factor for node with value: 210
Left child: Node(value=108, height=0, balance_factor=0), Right child: None
Updated height: 1
Updated balance factor: 1
Updating height and balance factor for node with value: 348

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Left child: Node(value=210, height=1, balance_factor=1), Right child: None
 Updated height: 2
 Updated balance factor: 2
 Inserted Node(value=210, height=1, balance_factor=1) as left child of
 Node(value=348, height=2, balance_factor=2)
 Right Rotate: k2_node=Node(value=348, height=2, balance_factor=2),
 root_node=Node(value=348, height=2, balance_factor=2)
 LL Case: k1_node=Node(value=210, height=1, balance_factor=1),
 k2_node=Node(value=348, height=2, balance_factor=2)
 Setting right child of node with value 210 to Node(value=348, height=2,
 balance_factor=2)
 Updating height and balance factor for node with value: 348
 Left child: Node(value=210, height=1, balance_factor=1), Right child: None
 Updated height: 2
 Updated balance factor: 2
 Updating height and balance factor for node with value: 210
 Left child: Node(value=108, height=0, balance_factor=0), Right child:
 Node(value=348, height=2, balance_factor=2)
 Updated height: 3
 Updated balance factor: -2
 Setting left child of node with value 348 to None
 Updating height and balance factor for node with value: 348
 Left child: None, Right child: None
 Updated height: 0
 Updated balance factor: 0
 Updating parent of None to Node(value=348, height=0, balance_factor=0)
 Setting parent of node with value 210 to Node(value=420, height=2,
 balance_factor=1)
 Updating height and balance factor for node with value: 420
 Left child: Node(value=348, height=0, balance_factor=0), Right child:
 Node(value=911, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: 0
 Setting left child of node with value 420 to Node(value=210, height=3,
 balance_factor=-2)
 Updating height and balance factor for node with value: 210
 Left child: Node(value=108, height=0, balance_factor=0), Right child:
 Node(value=348, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: 0
 Updating height and balance factor for node with value: 420
 Left child: Node(value=210, height=1, balance_factor=0), Right child:
 Node(value=911, height=0, balance_factor=0)
 Updated height: 2
 Updated balance factor: 1
 Updating height and balance factor for node with value: 420
 Left child: Node(value=210, height=1, balance_factor=0), Right child:
 Node(value=911, height=0, balance_factor=0)

Updated height: 2
 Updated balance factor: 1
 Updating height and balance factor for node with value: 210
 Left child: Node(value=108, height=0, balance_factor=0), Right child:
 Node(value=348, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: 0
 Updating height and balance factor for node with value: 348
 Left child: None, Right child: None
 Updated height: 0
 Updated balance factor: 0
 Updating height and balance factor for node with value: 420
 Left child: Node(value=210, height=1, balance_factor=0), Right child:
 Node(value=911, height=0, balance_factor=0)
 Updated height: 2
 Updated balance factor: 1
 Updating height and balance factor for node with value: 210
 Left child: Node(value=108, height=0, balance_factor=0), Right child:
 Node(value=348, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: 0
 Setting left child of node with value 420 to Node(value=210, height=1,
 balance_factor=0)
 Updating height and balance factor for node with value: 210
 Left child: Node(value=108, height=0, balance_factor=0), Right child:
 Node(value=348, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: 0
 Updating height and balance factor for node with value: 420
 Left child: Node(value=210, height=1, balance_factor=0), Right child:
 Node(value=911, height=0, balance_factor=0)
 Updated height: 2
 Updated balance factor: 1
 Inserted Node(value=210, height=1, balance_factor=0) as left child of
 Node(value=420, height=2, balance_factor=1)
 Setting right child of node with value 90 to Node(value=420, height=2,
 balance_factor=1)
 Updating height and balance factor for node with value: 420
 Left child: Node(value=210, height=1, balance_factor=0), Right child:
 Node(value=911, height=0, balance_factor=0)
 Updated height: 2
 Updated balance factor: 1
 Updating height and balance factor for node with value: 90
 Left child: Node(value=77, height=1, balance_factor=1), Right child:
 Node(value=420, height=2, balance_factor=1)
 Updated height: 3
 Updated balance factor: -1
 Inserted Node(value=420, height=2, balance_factor=1) as right child of

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Node(value=90, height=3, balance_factor=-1)
Setting right child of node with value 70 to Node(value=90, height=3,
balance_factor=-1)
Updating height and balance factor for node with value: 90
Left child: Node(value=77, height=1, balance_factor=1), Right child:
Node(value=420, height=2, balance_factor=1)
Updated height: 3
Updated balance factor: -1
Updating height and balance factor for node with value: 70
Left child: Node(value=55, height=2, balance_factor=0), Right child:
Node(value=90, height=3, balance_factor=-1)
Updated height: 4
Updated balance factor: -1
Inserted Node(value=90, height=3, balance_factor=-1) as right child of
Node(value=70, height=4, balance_factor=-1)
Created Node(value=768, height=None, balance_factor=None)
Setting left child of node with value 911 to Node(value=768, height=None,
balance_factor=None)
Updating height and balance factor for node with value: 768
Left child: None, Right child: None
Updated height: 0
Updated balance factor: 0
Updating height and balance factor for node with value: 911
Left child: Node(value=768, height=0, balance_factor=0), Right child: None
Updated height: 1
Updated balance factor: 1
Inserted Node(value=768, height=0, balance_factor=0) as left child of
Node(value=911, height=1, balance_factor=1)
Setting right child of node with value 420 to Node(value=911, height=1,
balance_factor=1)
Updating height and balance factor for node with value: 911
Left child: Node(value=768, height=0, balance_factor=0), Right child: None
Updated height: 1
Updated balance factor: 1
Updating height and balance factor for node with value: 420
Left child: Node(value=210, height=1, balance_factor=0), Right child:
Node(value=911, height=1, balance_factor=1)
Updated height: 2
Updated balance factor: 0
Inserted Node(value=911, height=1, balance_factor=1) as right child of
Node(value=420, height=2, balance_factor=0)
Setting right child of node with value 90 to Node(value=420, height=2,
balance_factor=0)
Updating height and balance factor for node with value: 420
Left child: Node(value=210, height=1, balance_factor=0), Right child:
Node(value=911, height=1, balance_factor=1)
Updated height: 2
Updated balance factor: 0

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Updating height and balance factor for node with value: 90
 Left child: Node(value=77, height=1, balance_factor=1), Right child:
 Node(value=420, height=2, balance_factor=0)
 Updated height: 3
 Updated balance factor: -1
 Inserted Node(value=420, height=2, balance_factor=0) as right child of
 Node(value=90, height=3, balance_factor=-1)
 Setting right child of node with value 70 to Node(value=90, height=3,
 balance_factor=-1)
 Updating height and balance factor for node with value: 90
 Left child: Node(value=77, height=1, balance_factor=1), Right child:
 Node(value=420, height=2, balance_factor=0)
 Updated height: 3
 Updated balance factor: -1
 Updating height and balance factor for node with value: 70
 Left child: Node(value=55, height=2, balance_factor=0), Right child:
 Node(value=90, height=3, balance_factor=-1)
 Updated height: 4
 Updated balance factor: -1
 Inserted Node(value=90, height=3, balance_factor=-1) as right child of
 Node(value=70, height=4, balance_factor=-1)
 Created Node(value=1080, height=None, balance_factor=None)
 Setting right child of node with value 911 to Node(value=1080, height=None,
 balance_factor=None)
 Updating height and balance factor for node with value: 1080
 Left child: None, Right child: None
 Updated height: 0
 Updated balance factor: 0
 Updating height and balance factor for node with value: 911
 Left child: Node(value=768, height=0, balance_factor=0), Right child:
 Node(value=1080, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: 0
 Inserted Node(value=1080, height=0, balance_factor=0) as right child of
 Node(value=911, height=1, balance_factor=0)
 Setting right child of node with value 420 to Node(value=911, height=1,
 balance_factor=0)
 Updating height and balance factor for node with value: 911
 Left child: Node(value=768, height=0, balance_factor=0), Right child:
 Node(value=1080, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: 0
 Updating height and balance factor for node with value: 420
 Left child: Node(value=210, height=1, balance_factor=0), Right child:
 Node(value=911, height=1, balance_factor=0)
 Updated height: 2
 Updated balance factor: 0
 Inserted Node(value=911, height=1, balance_factor=0) as right child of

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Node(value=420, height=2, balance_factor=0)
Setting right child of node with value 90 to Node(value=420, height=2,
balance_factor=0)
Updating height and balance factor for node with value: 420
Left child: Node(value=210, height=1, balance_factor=0), Right child:
Node(value=911, height=1, balance_factor=0)
Updated height: 2
Updated balance factor: 0
Updating height and balance factor for node with value: 90
Left child: Node(value=77, height=1, balance_factor=1), Right child:
Node(value=420, height=2, balance_factor=0)
Updated height: 3
Updated balance factor: -1
Inserted Node(value=420, height=2, balance_factor=0) as right child of
Node(value=90, height=3, balance_factor=-1)
Setting right child of node with value 70 to Node(value=90, height=3,
balance_factor=-1)
Updating height and balance factor for node with value: 90
Left child: Node(value=77, height=1, balance_factor=1), Right child:
Node(value=420, height=2, balance_factor=0)
Updated height: 3
Updated balance factor: -1
Updating height and balance factor for node with value: 70
Left child: Node(value=55, height=2, balance_factor=0), Right child:
Node(value=90, height=3, balance_factor=-1)
Updated height: 4
Updated balance factor: -1
Inserted Node(value=90, height=3, balance_factor=-1) as right child of
Node(value=70, height=4, balance_factor=-1)
Created Node(value=720, height=None, balance_factor=None)
Setting left child of node with value 768 to Node(value=720, height=None,
balance_factor=None)
Updating height and balance factor for node with value: 720
Left child: None, Right child: None
Updated height: 0
Updated balance factor: 0
Updating height and balance factor for node with value: 768
Left child: Node(value=720, height=0, balance_factor=0), Right child: None
Updated height: 1
Updated balance factor: 1
Inserted Node(value=720, height=0, balance_factor=0) as left child of
Node(value=768, height=1, balance_factor=1)
Setting left child of node with value 911 to Node(value=768, height=1,
balance_factor=1)
Updating height and balance factor for node with value: 768
Left child: Node(value=720, height=0, balance_factor=0), Right child: None
Updated height: 1
Updated balance factor: 1

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Updating height and balance factor for node with value: 911
 Left child: Node(value=768, height=1, balance_factor=1), Right child:
 Node(value=1080, height=0, balance_factor=0)
 Updated height: 2
 Updated balance factor: 1
 Inserted Node(value=768, height=1, balance_factor=1) as left child of
 Node(value=911, height=2, balance_factor=1)
 Setting right child of node with value 420 to Node(value=911, height=2,
 balance_factor=1)
 Updating height and balance factor for node with value: 911
 Left child: Node(value=768, height=1, balance_factor=1), Right child:
 Node(value=1080, height=0, balance_factor=0)
 Updated height: 2
 Updated balance factor: 1
 Updating height and balance factor for node with value: 420
 Left child: Node(value=210, height=1, balance_factor=0), Right child:
 Node(value=911, height=2, balance_factor=1)
 Updated height: 3
 Updated balance factor: -1
 Inserted Node(value=911, height=2, balance_factor=1) as right child of
 Node(value=420, height=3, balance_factor=-1)
 Setting right child of node with value 90 to Node(value=420, height=3,
 balance_factor=-1)
 Updating height and balance factor for node with value: 420
 Left child: Node(value=210, height=1, balance_factor=0), Right child:
 Node(value=911, height=2, balance_factor=1)
 Updated height: 3
 Updated balance factor: -1
 Updating height and balance factor for node with value: 90
 Left child: Node(value=77, height=1, balance_factor=1), Right child:
 Node(value=420, height=3, balance_factor=-1)
 Updated height: 4
 Updated balance factor: -2
 Inserted Node(value=420, height=3, balance_factor=-1) as right child of
 Node(value=90, height=4, balance_factor=-2)
 Left Rotate: k1_node=Node(value=90, height=4, balance_factor=-2),
 root_node=Node(value=90, height=4, balance_factor=-2)
 RR Case: k1_node=Node(value=90, height=4, balance_factor=-2),
 k2_node=Node(value=420, height=3, balance_factor=-1)
 Setting left child of node with value 420 to Node(value=90, height=4,
 balance_factor=-2)
 Updating height and balance factor for node with value: 90
 Left child: Node(value=77, height=1, balance_factor=1), Right child:
 Node(value=420, height=3, balance_factor=-1)
 Updated height: 4
 Updated balance factor: -2
 Updating height and balance factor for node with value: 420
 Left child: Node(value=90, height=4, balance_factor=-2), Right child:

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Node(value=911, height=2, balance_factor=1)
Updated height: 5
Updated balance factor: 2
Setting right child of node with value 90 to Node(value=210, height=1,
balance_factor=0)
Updating height and balance factor for node with value: 210
Left child: Node(value=108, height=0, balance_factor=0), Right child:
Node(value=348, height=0, balance_factor=0)
Updated height: 1
Updated balance factor: 0
Updating height and balance factor for node with value: 90
Left child: Node(value=77, height=1, balance_factor=1), Right child:
Node(value=210, height=1, balance_factor=0)
Updated height: 2
Updated balance factor: 0
Setting parent of node with value 210 to Node(value=90, height=2,
balance_factor=0)
Updating height and balance factor for node with value: 90
Left child: Node(value=77, height=1, balance_factor=1), Right child:
Node(value=210, height=1, balance_factor=0)
Updated height: 2
Updated balance factor: 0
Setting parent of node with value 420 to Node(value=70, height=4,
balance_factor=-1)
Updating height and balance factor for node with value: 70
Left child: Node(value=55, height=2, balance_factor=0), Right child:
Node(value=90, height=2, balance_factor=0)
Updated height: 3
Updated balance factor: 0
Setting right child of node with value 70 to Node(value=420, height=5,
balance_factor=2)
Updating height and balance factor for node with value: 420
Left child: Node(value=90, height=2, balance_factor=0), Right child:
Node(value=911, height=2, balance_factor=1)
Updated height: 3
Updated balance factor: 0
Updating height and balance factor for node with value: 70
Left child: Node(value=55, height=2, balance_factor=0), Right child:
Node(value=420, height=3, balance_factor=0)
Updated height: 4
Updated balance factor: -1
Updating height and balance factor for node with value: 70
Left child: Node(value=55, height=2, balance_factor=0), Right child:
Node(value=420, height=3, balance_factor=0)
Updated height: 4
Updated balance factor: -1
Updating height and balance factor for node with value: 90
Left child: Node(value=77, height=1, balance_factor=1), Right child:

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Node(value=210, height=1, balance_factor=0)
Updated height: 2
Updated balance factor: 0
Updating height and balance factor for node with value: 420
Left child: Node(value=90, height=2, balance_factor=0), Right child:
Node(value=911, height=2, balance_factor=1)
Updated height: 3
Updated balance factor: 0
Updating height and balance factor for node with value: 70
Left child: Node(value=55, height=2, balance_factor=0), Right child:
Node(value=420, height=3, balance_factor=0)
Updated height: 4
Updated balance factor: -1
Updating height and balance factor for node with value: 420
Left child: Node(value=90, height=2, balance_factor=0), Right child:
Node(value=911, height=2, balance_factor=1)
Updated height: 3
Updated balance factor: 0
Setting right child of node with value 70 to Node(value=420, height=3,
balance_factor=0)
Updating height and balance factor for node with value: 420
Left child: Node(value=90, height=2, balance_factor=0), Right child:
Node(value=911, height=2, balance_factor=1)
Updated height: 3
Updated balance factor: 0
Updating height and balance factor for node with value: 70
Left child: Node(value=55, height=2, balance_factor=0), Right child:
Node(value=420, height=3, balance_factor=0)
Updated height: 4
Updated balance factor: -1
Inserted Node(value=420, height=3, balance_factor=0) as right child of
Node(value=70, height=4, balance_factor=-1)
Updated balance factor: 2
Setting right child of node with value 348 to None
Updating height and balance factor for node with value: 348
Left child: None, Right child: None
Updated height: 0
Updated balance factor: 0
Setting parent of node with value 420 to Node(value=90, height=2,
balance_factor=0)
Updating height and balance factor for node with value: 90
Left child: Node(value=77, height=1, balance_factor=1), Right child:
Node(value=348, height=0, balance_factor=0)
Updated height: 2
Updated balance factor: 1
Setting right child of node with value 90 to Node(value=420, height=3,
balance_factor=2)
Updating height and balance factor for node with value: 420

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Left child: Node(value=348, height=0, balance_factor=0), Right child:
 Node(value=911, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: 0
 Updating height and balance factor for node with value: 90
 Left child: Node(value=77, height=1, balance_factor=1), Right child:
 Node(value=420, height=1, balance_factor=0)
 Updated height: 2
 Updated balance factor: 0
 Updating height and balance factor for node with value: 90
 Left child: Node(value=77, height=1, balance_factor=1), Right child:
 Node(value=420, height=1, balance_factor=0)
 Updated height: 2
 Updated balance factor: 0
 Updating height and balance factor for node with value: 348
 Left child: None, Right child: None
 Updated height: 0
 Updated balance factor: 0
 Updating height and balance factor for node with value: 420
 Left child: Node(value=348, height=0, balance_factor=0), Right child:
 Node(value=911, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: 0
 Updating height and balance factor for node with value: 90
 Left child: Node(value=77, height=1, balance_factor=1), Right child:
 Node(value=420, height=1, balance_factor=0)
 Updated height: 2
 Updated balance factor: 0
 Updating height and balance factor for node with value: 420
 Left child: Node(value=348, height=0, balance_factor=0), Right child:
 Node(value=911, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: 0
 Setting right child of node with value 90 to Node(value=420, height=1,
 balance_factor=0)
 Updating height and balance factor for node with value: 420
 Left child: Node(value=348, height=0, balance_factor=0), Right child:
 Node(value=911, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: 0
 Updating height and balance factor for node with value: 90
 Left child: Node(value=77, height=1, balance_factor=1), Right child:
 Node(value=420, height=1, balance_factor=0)
 Updated height: 2
 Updated balance factor: 0
 Inserted Node(value=420, height=1, balance_factor=0) as right child of
 Node(value=90, height=2, balance_factor=0)
 Setting right child of node with value 70 to Node(value=90, height=2,

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balance_factor=0)
Updating height and balance factor for node with value: 90
Left child: Node(value=77, height=1, balance_factor=1), Right child:
Node(value=420, height=1, balance_factor=0)
Updated height: 2
Updated balance factor: 0
Updating height and balance factor for node with value: 70
Left child: Node(value=55, height=2, balance_factor=0), Right child:
Node(value=90, height=2, balance_factor=0)
Updated height: 3
Updated balance factor: 0
Inserted Node(value=90, height=2, balance_factor=0) as right child of
Node(value=70, height=3, balance_factor=0)
Created Node(value=210, height=None, balance_factor=None)
Setting left child of node with value 348 to Node(value=210, height=None,
balance_factor=None)
Updating height and balance factor for node with value: 210
Left child: None, Right child: None
Updated height: 0
Updated balance factor: 0
Updating height and balance factor for node with value: 348
Left child: Node(value=210, height=0, balance_factor=0), Right child: None
Updated height: 1
Updated balance factor: 1
Inserted Node(value=210, height=0, balance_factor=0) as left child of
Node(value=348, height=1, balance_factor=1)
Setting left child of node with value 420 to Node(value=348, height=1,
balance_factor=1)
Updating height and balance factor for node with value: 348
Left child: Node(value=210, height=0, balance_factor=0), Right child: None
Updated height: 1
Updated balance factor: 1
Updating height and balance factor for node with value: 420
Left child: Node(value=348, height=1, balance_factor=1), Right child:
Node(value=911, height=0, balance_factor=0)
Updated height: 2
Updated balance factor: 1
Inserted Node(value=348, height=1, balance_factor=1) as left child of
Node(value=420, height=2, balance_factor=1)
Setting right child of node with value 90 to Node(value=420, height=2,
balance_factor=1)
Updating height and balance factor for node with value: 420
Left child: Node(value=348, height=1, balance_factor=1), Right child:
Node(value=911, height=0, balance_factor=0)
Updated height: 2
Updated balance factor: 1
Updating height and balance factor for node with value: 90
Left child: Node(value=77, height=1, balance_factor=1), Right child:

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Node(value=420, height=2, balance_factor=1)
Updated height: 3
Updated balance factor: -1
Inserted Node(value=420, height=2, balance_factor=1) as right child of
Node(value=90, height=3, balance_factor=-1)
Setting right child of node with value 70 to Node(value=90, height=3,
balance_factor=-1)
Updating height and balance factor for node with value: 90
Left child: Node(value=77, height=1, balance_factor=1), Right child:
Node(value=420, height=2, balance_factor=1)
Updated height: 3
Updated balance factor: -1
Updating height and balance factor for node with value: 70
Left child: Node(value=55, height=2, balance_factor=0), Right child:
Node(value=90, height=3, balance_factor=-1)
Updated height: 4
Updated balance factor: -1
Inserted Node(value=90, height=3, balance_factor=-1) as right child of
Node(value=70, height=4, balance_factor=-1)
Created Node(value=108, height=None, balance_factor=None)
Setting left child of node with value 210 to Node(value=108, height=None,
balance_factor=None)
Updating height and balance factor for node with value: 108
Left child: None, Right child: None
Updated height: 0
Updated balance factor: 0
Updating height and balance factor for node with value: 210
Left child: Node(value=108, height=0, balance_factor=0), Right child: None
Updated height: 1
Updated balance factor: 1
Inserted Node(value=108, height=0, balance_factor=0) as left child of
Node(value=210, height=1, balance_factor=1)
Setting left child of node with value 348 to Node(value=210, height=1,
balance_factor=1)
Updating height and balance factor for node with value: 210
Left child: Node(value=108, height=0, balance_factor=0), Right child: None
Updated height: 1
Updated balance factor: 1
Updating height and balance factor for node with value: 348
Left child: Node(value=210, height=1, balance_factor=1), Right child: None
Updated height: 2
Updated balance factor: 2
Inserted Node(value=210, height=1, balance_factor=1) as left child of
Node(value=348, height=2, balance_factor=2)
Right Rotate: k2_node=Node(value=348, height=2, balance_factor=2),
root_node=Node(value=348, height=2, balance_factor=2)
LL Case: k1_node=Node(value=210, height=1, balance_factor=1),
k2_node=Node(value=348, height=2, balance_factor=2)

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Setting right child of node with value 210 to Node(value=348, height=2, balance_factor=2)
 Updating height and balance factor for node with value: 348
 Left child: Node(value=210, height=1, balance_factor=1), Right child: None
 Updated height: 2
 Updated balance factor: 2
 Updating height and balance factor for node with value: 210
 Left child: Node(value=108, height=0, balance_factor=0), Right child: Node(value=348, height=2, balance_factor=2)
 Updated height: 3
 Updated balance factor: -2
 Setting left child of node with value 348 to None
 Updating height and balance factor for node with value: 348
 Left child: None, Right child: None
 Updated height: 0
 Updated balance factor: 0
 Updating parent of None to Node(value=348, height=0, balance_factor=0)
 Setting parent of node with value 210 to Node(value=420, height=2, balance_factor=1)
 Updating height and balance factor for node with value: 420
 Left child: Node(value=348, height=0, balance_factor=0), Right child: Node(value=911, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: 0
 Setting left child of node with value 420 to Node(value=210, height=3, balance_factor=-2)
 Updating height and balance factor for node with value: 210
 Left child: Node(value=108, height=0, balance_factor=0), Right child: Node(value=348, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: 0
 Updating height and balance factor for node with value: 420
 Left child: Node(value=210, height=1, balance_factor=0), Right child: Node(value=911, height=0, balance_factor=0)
 Updated height: 2
 Updated balance factor: 1
 Updating height and balance factor for node with value: 420
 Left child: Node(value=210, height=1, balance_factor=0), Right child: Node(value=911, height=0, balance_factor=0)
 Updated height: 2
 Updated balance factor: 1
 Updating height and balance factor for node with value: 210
 Left child: Node(value=108, height=0, balance_factor=0), Right child: Node(value=348, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: 0
 Updating height and balance factor for node with value: 348
 Left child: None, Right child: None

Updated height: 0
 Updated balance factor: 0
 Updating height and balance factor for node with value: 420
 Left child: Node(value=210, height=1, balance_factor=0), Right child:
 Node(value=911, height=0, balance_factor=0)
 Updated height: 2
 Updated balance factor: 1
 Updating height and balance factor for node with value: 210
 Left child: Node(value=108, height=0, balance_factor=0), Right child:
 Node(value=348, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: 0
 Setting left child of node with value 420 to Node(value=210, height=1,
 balance_factor=0)
 Updating height and balance factor for node with value: 210
 Left child: Node(value=108, height=0, balance_factor=0), Right child:
 Node(value=348, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: 0
 Updating height and balance factor for node with value: 420
 Left child: Node(value=210, height=1, balance_factor=0), Right child:
 Node(value=911, height=0, balance_factor=0)
 Updated height: 2
 Updated balance factor: 1
 Inserted Node(value=210, height=1, balance_factor=0) as left child of
 Node(value=420, height=2, balance_factor=1)
 Setting right child of node with value 90 to Node(value=420, height=2,
 balance_factor=1)
 Updating height and balance factor for node with value: 420
 Left child: Node(value=210, height=1, balance_factor=0), Right child:
 Node(value=911, height=0, balance_factor=0)
 Updated height: 2
 Updated balance factor: 1
 Updating height and balance factor for node with value: 90
 Left child: Node(value=77, height=1, balance_factor=1), Right child:
 Node(value=420, height=2, balance_factor=1)
 Updated height: 3
 Updated balance factor: -1
 Inserted Node(value=420, height=2, balance_factor=1) as right child of
 Node(value=90, height=3, balance_factor=-1)
 Setting right child of node with value 70 to Node(value=90, height=3,
 balance_factor=-1)
 Updating height and balance factor for node with value: 90
 Left child: Node(value=77, height=1, balance_factor=1), Right child:
 Node(value=420, height=2, balance_factor=1)
 Updated height: 3
 Updated balance factor: -1
 Updating height and balance factor for node with value: 70

Left child: Node(value=55, height=2, balance_factor=0), Right child:
 Node(value=90, height=3, balance_factor=-1)
 Updated height: 4
 Updated balance factor: -1
 Inserted Node(value=90, height=3, balance_factor=-1) as right child of
 Node(value=70, height=4, balance_factor=-1)
 Created Node(value=768, height=None, balance_factor=None)
 Setting left child of node with value 911 to Node(value=768, height=None,
 balance_factor=None)
 Updating height and balance factor for node with value: 768
 Left child: None, Right child: None
 Updated height: 0
 Updated balance factor: 0
 Updating height and balance factor for node with value: 911
 Left child: Node(value=768, height=0, balance_factor=0), Right child: None
 Updated height: 1
 Updated balance factor: 1
 Inserted Node(value=768, height=0, balance_factor=0) as left child of
 Node(value=911, height=1, balance_factor=1)
 Setting right child of node with value 420 to Node(value=911, height=1,
 balance_factor=1)
 Updating height and balance factor for node with value: 911
 Left child: Node(value=768, height=0, balance_factor=0), Right child: None
 Updated height: 1
 Updated balance factor: 1
 Updating height and balance factor for node with value: 420
 Left child: Node(value=210, height=1, balance_factor=0), Right child:
 Node(value=911, height=1, balance_factor=1)
 Updated height: 2
 Updated balance factor: 0
 Inserted Node(value=911, height=1, balance_factor=1) as right child of
 Node(value=420, height=2, balance_factor=0)
 Setting right child of node with value 90 to Node(value=420, height=2,
 balance_factor=0)
 Updating height and balance factor for node with value: 420
 Left child: Node(value=210, height=1, balance_factor=0), Right child:
 Node(value=911, height=1, balance_factor=1)
 Updated height: 2
 Updated balance factor: 0
 Updating height and balance factor for node with value: 90
 Left child: Node(value=77, height=1, balance_factor=1), Right child:
 Node(value=420, height=2, balance_factor=0)
 Updated height: 3
 Updated balance factor: -1
 Inserted Node(value=420, height=2, balance_factor=0) as right child of
 Node(value=90, height=3, balance_factor=-1)
 Setting right child of node with value 70 to Node(value=90, height=3,
 balance_factor=-1)

Updating height and balance factor for node with value: 90
 Left child: Node(value=77, height=1, balance_factor=1), Right child:
 Node(value=420, height=2, balance_factor=0)
 Updated height: 3
 Updated balance factor: -1
 Updating height and balance factor for node with value: 70
 Left child: Node(value=55, height=2, balance_factor=0), Right child:
 Node(value=90, height=3, balance_factor=-1)
 Updated height: 4
 Updated balance factor: -1
 Inserted Node(value=90, height=3, balance_factor=-1) as right child of
 Node(value=70, height=4, balance_factor=-1)
 Created Node(value=1080, height=None, balance_factor=None)
 Setting right child of node with value 911 to Node(value=1080, height=None,
 balance_factor=None)
 Updating height and balance factor for node with value: 1080
 Left child: None, Right child: None
 Updated height: 0
 Updated balance factor: 0
 Updating height and balance factor for node with value: 911
 Left child: Node(value=768, height=0, balance_factor=0), Right child:
 Node(value=1080, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: 0
 Inserted Node(value=1080, height=0, balance_factor=0) as right child of
 Node(value=911, height=1, balance_factor=0)
 Setting right child of node with value 420 to Node(value=911, height=1,
 balance_factor=0)
 Updating height and balance factor for node with value: 911
 Left child: Node(value=768, height=0, balance_factor=0), Right child:
 Node(value=1080, height=0, balance_factor=0)
 Updated height: 1
 Updated balance factor: 0
 Updating height and balance factor for node with value: 420
 Left child: Node(value=210, height=1, balance_factor=0), Right child:
 Node(value=911, height=1, balance_factor=0)
 Updated height: 2
 Updated balance factor: 0
 Inserted Node(value=911, height=1, balance_factor=0) as right child of
 Node(value=420, height=2, balance_factor=0)
 Setting right child of node with value 90 to Node(value=420, height=2,
 balance_factor=0)
 Updating height and balance factor for node with value: 420
 Left child: Node(value=210, height=1, balance_factor=0), Right child:
 Node(value=911, height=1, balance_factor=0)
 Updated height: 2
 Updated balance factor: 0
 Updating height and balance factor for node with value: 90

Left child: Node(value=77, height=1, balance_factor=1), Right child:
 Node(value=420, height=2, balance_factor=0)
 Updated height: 3
 Updated balance factor: -1
 Inserted Node(value=420, height=2, balance_factor=0) as right child of
 Node(value=90, height=3, balance_factor=-1)
 Setting right child of node with value 70 to Node(value=90, height=3,
 balance_factor=-1)
 Updating height and balance factor for node with value: 90
 Left child: Node(value=77, height=1, balance_factor=1), Right child:
 Node(value=420, height=2, balance_factor=0)
 Updated height: 3
 Updated balance factor: -1
 Updating height and balance factor for node with value: 70
 Left child: Node(value=55, height=2, balance_factor=0), Right child:
 Node(value=90, height=3, balance_factor=-1)
 Updated height: 4
 Updated balance factor: -1
 Inserted Node(value=90, height=3, balance_factor=-1) as right child of
 Node(value=70, height=4, balance_factor=-1)
 Created Node(value=720, height=None, balance_factor=None)
 Setting left child of node with value 768 to Node(value=720, height=None,
 balance_factor=None)
 Updating height and balance factor for node with value: 720
 Left child: None, Right child: None
 Updated height: 0
 Updated balance factor: 0
 Updating height and balance factor for node with value: 768
 Left child: Node(value=720, height=0, balance_factor=0), Right child: None
 Updated height: 1
 Updated balance factor: 1
 Inserted Node(value=720, height=0, balance_factor=0) as left child of
 Node(value=768, height=1, balance_factor=1)
 Setting left child of node with value 911 to Node(value=768, height=1,
 balance_factor=1)
 Updating height and balance factor for node with value: 768
 Left child: Node(value=720, height=0, balance_factor=0), Right child: None
 Updated height: 1
 Updated balance factor: 1
 Updating height and balance factor for node with value: 911
 Left child: Node(value=768, height=1, balance_factor=1), Right child:
 Node(value=1080, height=0, balance_factor=0)
 Updated height: 2
 Updated balance factor: 1
 Inserted Node(value=768, height=1, balance_factor=1) as left child of
 Node(value=911, height=2, balance_factor=1)
 Setting right child of node with value 420 to Node(value=911, height=2,
 balance_factor=1)

Updating height and balance factor for node with value: 911
 Left child: Node(value=768, height=1, balance_factor=1), Right child:
 Node(value=1080, height=0, balance_factor=0)
 Updated height: 2
 Updated balance factor: 1
 Updating height and balance factor for node with value: 420
 Left child: Node(value=210, height=1, balance_factor=0), Right child:
 Node(value=911, height=2, balance_factor=1)
 Updated height: 3
 Updated balance factor: -1
 Inserted Node(value=911, height=2, balance_factor=1) as right child of
 Node(value=420, height=3, balance_factor=-1)
 Setting right child of node with value 90 to Node(value=420, height=3,
 balance_factor=-1)
 Updating height and balance factor for node with value: 420
 Left child: Node(value=210, height=1, balance_factor=0), Right child:
 Node(value=911, height=2, balance_factor=1)
 Updated height: 3
 Updated balance factor: -1
 Updating height and balance factor for node with value: 90
 Left child: Node(value=77, height=1, balance_factor=1), Right child:
 Node(value=420, height=3, balance_factor=-1)
 Updated height: 4
 Updated balance factor: -2
 Inserted Node(value=420, height=3, balance_factor=-1) as right child of
 Node(value=90, height=4, balance_factor=-2)
 Left Rotate: k1_node=Node(value=90, height=4, balance_factor=-2),
 root_node=Node(value=90, height=4, balance_factor=-2)
 RR Case: k1_node=Node(value=90, height=4, balance_factor=-2),
 k2_node=Node(value=420, height=3, balance_factor=-1)
 Setting left child of node with value 420 to Node(value=90, height=4,
 balance_factor=-2)
 Updating height and balance factor for node with value: 90
 Left child: Node(value=77, height=1, balance_factor=1), Right child:
 Node(value=420, height=3, balance_factor=-1)
 Updated height: 4
 Updated balance factor: -2
 Updating height and balance factor for node with value: 420
 Left child: Node(value=90, height=4, balance_factor=-2), Right child:
 Node(value=911, height=2, balance_factor=1)
 Updated height: 5
 Updated balance factor: 2
 Setting right child of node with value 90 to Node(value=210, height=1,
 balance_factor=0)
 Updating height and balance factor for node with value: 210
 Left child: Node(value=108, height=0, balance_factor=0), Right child:
 Node(value=348, height=0, balance_factor=0)
 Updated height: 1

Updated balance factor: 0
 Updating height and balance factor for node with value: 90
 Left child: Node(value=77, height=1, balance_factor=1), Right child:
 Node(value=210, height=1, balance_factor=0)
 Updated height: 2
 Updated balance factor: 0
 Setting parent of node with value 210 to Node(value=90, height=2,
 balance_factor=0)
 Updating height and balance factor for node with value: 90
 Left child: Node(value=77, height=1, balance_factor=1), Right child:
 Node(value=210, height=1, balance_factor=0)
 Updated height: 2
 Updated balance factor: 0
 Setting parent of node with value 420 to Node(value=70, height=4,
 balance_factor=-1)
 Updating height and balance factor for node with value: 70
 Left child: Node(value=55, height=2, balance_factor=0), Right child:
 Node(value=90, height=2, balance_factor=0)
 Updated height: 3
 Updated balance factor: 0
 Setting right child of node with value 70 to Node(value=420, height=5,
 balance_factor=2)
 Updating height and balance factor for node with value: 420
 Left child: Node(value=90, height=2, balance_factor=0), Right child:
 Node(value=911, height=2, balance_factor=1)
 Updated height: 3
 Updated balance factor: 0
 Updating height and balance factor for node with value: 70
 Left child: Node(value=55, height=2, balance_factor=0), Right child:
 Node(value=420, height=3, balance_factor=0)
 Updated height: 4
 Updated balance factor: -1
 Updating height and balance factor for node with value: 70
 Left child: Node(value=55, height=2, balance_factor=0), Right child:
 Node(value=420, height=3, balance_factor=0)
 Updated height: 4
 Updated balance factor: -1
 Updating height and balance factor for node with value: 90
 Left child: Node(value=77, height=1, balance_factor=1), Right child:
 Node(value=210, height=1, balance_factor=0)
 Updated height: 2
 Updated balance factor: 0
 Updating height and balance factor for node with value: 420
 Left child: Node(value=90, height=2, balance_factor=0), Right child:
 Node(value=911, height=2, balance_factor=1)
 Updated height: 3
 Updated balance factor: 0
 Updating height and balance factor for node with value: 70

```

Left child: Node(value=55, height=2, balance_factor=0), Right child:
Node(value=420, height=3, balance_factor=0)
Updated height: 4
Updated balance factor: -1
Updating height and balance factor for node with value: 420
Left child: Node(value=90, height=2, balance_factor=0), Right child:
Node(value=911, height=2, balance_factor=1)
Updated height: 3
Updated balance factor: 0
Setting right child of node with value 70 to Node(value=420, height=3,
balance_factor=0)
Updating height and balance factor for node with value: 420
Left child: Node(value=90, height=2, balance_factor=0), Right child:
Node(value=911, height=2, balance_factor=1)
Updated height: 3
Updated balance factor: 0
Updating height and balance factor for node with value: 70
Left child: Node(value=55, height=2, balance_factor=0), Right child:
Node(value=420, height=3, balance_factor=0)
Updated height: 4
Updated balance factor: -1
Inserted Node(value=420, height=3, balance_factor=0) as right child of
Node(value=70, height=4, balance_factor=-1)

```

3 Q4

Perform the in-order traversal of the final AVL tree and write the traversal output to a new file (output.txt).

```

[6]: def AVL_inorder_traversal(root: AVL_Node | None, out_file: TextIOWrapper |
↳None) -> None:
    if root is None:
        return
    AVL_inorder_traversal(root.lchild, out_file = out_file)
    if out_file is not None:
        print(root.value, end = ', ', file = out_file)
    print(root.value, end = ', ')
    print(root)
    AVL_inorder_traversal(root.rchild, out_file = out_file)

def AVL_preorder_traversal(root: AVL_Node | None, out_file: TextIOWrapper |
↳None) -> None:
    if root is None:
        return
    if out_file is not None:
        print(root.value, end = ', ', file = out_file)
    print(root.value, end = ', ')

```

```

print(root)
AVL_preorder_traversal(root.lchild, out_file)
AVL_preorder_traversal(root.rchild, out_file)

def AVL_postorder_traversal(root: AVL_Node | None, out_file: TextIOWrapper | None) -> None:
    if root is None:
        return
    AVL_postorder_traversal(root.lchild, out_file)
    AVL_postorder_traversal(root.rchild, out_file)
    if out_file is not None:
        print(root.value, end = ', ', file = out_file)
    print(root.value, end=', ')
    print(root)

```

```

[7]: data1 = open("./2650026_lab3_data/output.txt", 'w')
      AVL_inorder_traversal(root0, data1)
      data1.close()

```

```

25, Node(value=25, height=0, balance_factor=0)
50, Node(value=50, height=1, balance_factor=1)
55, Node(value=55, height=2, balance_factor=0)
67, Node(value=67, height=1, balance_factor=-1)
69, Node(value=69, height=0, balance_factor=0)
70, Node(value=70, height=4, balance_factor=-1)
76, Node(value=76, height=0, balance_factor=0)
77, Node(value=77, height=1, balance_factor=1)
90, Node(value=90, height=2, balance_factor=0)
108, Node(value=108, height=0, balance_factor=0)
210, Node(value=210, height=1, balance_factor=0)
348, Node(value=348, height=0, balance_factor=0)
420, Node(value=420, height=3, balance_factor=0)
720, Node(value=720, height=0, balance_factor=0)
768, Node(value=768, height=1, balance_factor=1)
911, Node(value=911, height=2, balance_factor=1)
1080, Node(value=1080, height=0, balance_factor=0)

```

4 Q5

Display the height and balance factor of the tree before and after each operation (optional enhancement).

Answer:

Already applied in above code, refer above outputs.

5 Q6

Plot Height vs number of node

```
[8]: height_store = [] # Store the height of the root node after each insertion
root1 = AVL_Node() # Reset the root for the new insertions
arr1 = []
with open("./2650026_lab3_data/input0.txt", 'r') as data0:
    for line in data0.readlines():
        line_data = line.rstrip().rstrip(',').replace(' ', '').split(',')
        for data in line_data:
            arr1.append(int(data))

# print(arr1)
```

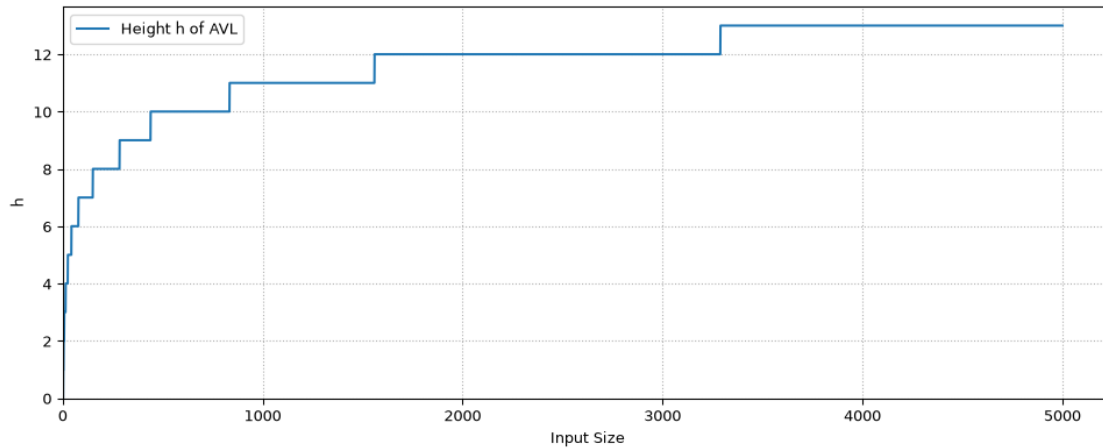
```
[9]: for i in arr1:
    root1 = AVL_insert(root1, i)
    # print(i)
    # print("height: ", root1.height if root1.height is not None else 0) # type:
    ↪ ignore
    height_store.append(root1.height if root1.height is not None else 0) # type:
    ↪ ignore
```

```
[10]: height_store_file = open("./2650026_lab3_data/height_store.txt", 'w')
for height in height_store:
    height_store_file.write(f"{height}, ")
height_store_file.write("\n")
height_store_file.close()
```

```
[11]: # Plotting the height of the AVL tree after each insertion
figure(figsize=(12, 4.5), dpi=97)

y = height_store[:]
x = [i+1 for i in range(len(y))]

# plt.subplot(3, 1, 2)
plt.plot(x, y, label="Height h of AVL")
plt.xlim(left=0)
plt.ylim(bottom=0)
plt.legend()
plt.xlabel("Input Size")
plt.ylabel("h")
plt.grid(visible=True, linestyle=':')
plt.show()
```



6 Q7

Compare AVL and BST search time

6.1 BST Tree: All Code

```
[12]: iterations = 0 # Global Variable to count iterations during a run
```

```
[13]: class BST_Node:
    def __init__(self):
        self.lchild = None
        self.rchild = None
        self.value = None
        self.parent = None

def BST_Insert(value, root_node:BST_Node | None = None) -> BST_Node:
    assert value != None, "Value cannot be None"
    y = None
    x = root_node
    global iterations
    while x is not None:
        iterations += 1
        y = x
        if value == x.value: # type: ignore
            if PRINT_DEBUG:
                print(f"Value {value} already exists in the BST. Duplicate_
↪values are not allowed.")
            return root_node # type: ignore
        elif value < x.value: # type: ignore
            x = x.lchild # type: ignore
        else:
```

```

        x = x.rchild # type: ignore
    new_node = BST_Node()
    new_node.value = value
    new_node.parent = y # type: ignore
    if y is None:
        root_node = new_node
    elif value < y.value: # type: ignore
        y.lchild = new_node # type: ignore
    else:
        y.rchild = new_node # type: ignore
    return root_node # type: ignore

def BST_inorder(root_node:BST_Node | None = None):
    global iterations
    if root_node is None:
        return
    iterations += 1 # type: ignore
    BST_inorder(root_node.lchild)
    print(root_node.value, end=',')
    BST_inorder(root_node.rchild)

def BST_preorder(root_node:BST_Node | None = None):
    global iterations
    if root_node is None:
        return
    iterations += 1 # type: ignore
    print(root_node.value, end=',')
    BST_preorder(root_node.lchild)
    BST_preorder(root_node.rchild)

def BST_postorder(root_node:BST_Node | None = None):
    global iterations
    if root_node is None:
        return
    iterations += 1 # type: ignore
    BST_postorder(root_node.lchild)
    BST_postorder(root_node.rchild)
    print(root_node.value, end=',')

def BST_Search(value, root_node:BST_Node | None = None) -> BST_Node | None:
    global iterations
    if root_node is None:
        return None
    iterations += 1 # type: ignore
    if value == root_node.value:
        return root_node
    elif value < root_node.value:

```

```

        return BST_Search(value, root_node.lchild)
    else:
        return BST_Search(value, root_node.rchild)

```

6.2 AVL Tree Searching Code

```

[14]: def AVL_Search(value, root_node: AVL_Node | None = None) -> AVL_Node | None:
    global iterations
    if root_node is None or root_node.value is None:
        return None
    iterations += 1 # type: ignore
    if value == root_node.value:
        return root_node
    elif value < root_node.value:
        return AVL_Search(value, root_node.lchild)
    else:
        return AVL_Search(value, root_node.rchild)

```

6.3 Compare iterations for AVL_Search vs BST_Search

```

[15]: def _deepest_avl_value(node: AVL_Node) -> int:
    # Cached height lets us walk straight to the deepest leaf in O(log n)
    while node.lchild is not None or node.rchild is not None:
        if node.lchild is None:
            node = node.rchild
        elif node.rchild is None:
            node = node.lchild
        elif node.lchild.height >= node.rchild.height:
            node = node.lchild
        else:
            node = node.rchild
    return node.value # type: ignore

def _median_insertion_order(sorted_vals: list) -> list:
    # Median-first insertion forces the most balanced BST shape possible
    if not sorted_vals:
        return []
    mid = len(sorted_vals) // 2
    return (
        [sorted_vals[mid]]
        + _median_insertion_order(sorted_vals[:mid])
        + _median_insertion_order(sorted_vals[mid + 1:])
    )

def compare_search_iterations(n: int, arr1: list) -> tuple:

```

```

global iterations

assert isinstance(n, int) and n > 0, "n must be a positive integer"
assert len(arr1) >= 5000, "arr1 must contain at least 5000 elements"
assert n <= len(arr1), "n cannot exceed the length of arr1"

start_idx = random.randint(0, len(arr1) - n)
elements = arr1[start_idx : start_idx + n]

# Main trees, built in sampled (effectively random) order
bst_root = None
for val in elements:
    bst_root = BST_Insert(val, bst_root)

avl_root = None
for val in elements:
    avl_root = AVL_insert(avl_root, val)

# Sorted-order insertion forces the true BST worst case (degenerate chain)
sorted_elements = sorted(set(elements))
worst_bst_root = None
for val in sorted_elements:
    worst_bst_root = BST_Insert(val, worst_bst_root)
worst_bst_value = sorted_elements[-1]

# Median-first insertion forces the most balanced BST possible (for
↪reference)
balanced_order = _median_insertion_order(sorted_elements)
balanced_bst_root = None
for val in balanced_order:
    balanced_bst_root = BST_Insert(val, balanced_bst_root)

# Best case: root is always depth 0, regardless of insertion order
best_bst_value = bst_root.value # type: ignore
best_avl_value = avl_root.value # type: ignore

# AVL worst case: rotations bound height to  $O(\log n)$  regardless of order
worst_avl_value = _deepest_avl_value(avl_root) # type: ignore

# Random case: any element actually inserted into the main trees
random_bst_value = random.choice(elements)
random_avl_value = random.choice(elements)

iterations = 0
BST_Search(best_bst_value, bst_root)
best_case_bst = iterations

```

```

iterations = 0
AVL_Search(best_avl_value, avl_root)
best_case_avl = iterations

iterations = 0
BST_Search(worst_bst_value, worst_bst_root) # searched in the sorted-order
↪ tree
worst_case_bst = iterations

iterations = 0
AVL_Search(worst_avl_value, avl_root)
worst_case_avl = iterations

iterations = 0
BST_Search(random_bst_value, bst_root)
random_case_bst = iterations

iterations = 0
AVL_Search(random_avl_value, avl_root)
random_case_avl = iterations

return (
    best_case_bst, best_case_avl,
    worst_case_bst, worst_case_avl,
    random_case_bst, random_case_avl,
)

```

```

[16]: import statistics

results = {}
trials = 10 # adjust as needed

for n in [10, 25, 50, 100, 250, 500, 1000, 2500, 5000]:
    runs = [compare_search_iterations(n, arr1) for _ in range(trials)]

    best_case_bst = min(run[0] for run in runs)
    best_case_avl = min(run[1] for run in runs)
    worst_case_bst = max(run[2] for run in runs)
    worst_case_avl = max(run[3] for run in runs)
    random_case_bst = statistics.mean(run[4] for run in runs)
    random_case_avl = statistics.mean(run[5] for run in runs)

    results[n] = (
        best_case_bst, best_case_avl,
        worst_case_bst, worst_case_avl,
        random_case_bst, random_case_avl,
    )

```

```
print("Results:\n", results)
```

Results:

```
{10: (1, 1, 10, 4, 3, 2.8), 25: (1, 1, 25, 6, 5.3, 4.6), 50: (1, 1, 50, 7, 5.4, 5), 100: (1, 1, 100, 8, 6.4, 6.2), 250: (1, 1, 250, 10, 8.4, 7.5), 500: (1, 1, 498, 11, 9.5, 9.2), 1000: (1, 1, 987, 12, 14.8, 8.8), 2500: (1, 1, 2409, 14, 33.9, 10.5), 5000: (1, 1, 4534, 14, 71.5, 11.8)}
```

```
[17]: search_columns = [
        "best_case_bst", "best_case_avl",
        "worst_case_bst", "worst_case_avl",
        "random_case_bst", "random_case_avl",
    ]

    search_df = pd.DataFrame.from_dict(results, orient="index",
        columns=search_columns)
    search_df.index.name = "n"
    search_df = search_df.reset_index()
```

```
[18]: search_df
```

```
[18]:
```

	n	best_case_bst	best_case_avl	worst_case_bst	worst_case_avl	\
0	10	1	1	10	4	
1	25	1	1	25	6	
2	50	1	1	50	7	
3	100	1	1	100	8	
4	250	1	1	250	10	
5	500	1	1	498	11	
6	1000	1	1	987	12	
7	2500	1	1	2409	14	
8	5000	1	1	4534	14	

	random_case_bst	random_case_avl
0	3.0	2.8
1	5.3	4.6
2	5.4	5.0
3	6.4	6.2
4	8.4	7.5
5	9.5	9.2
6	14.8	8.8
7	33.9	10.5
8	71.5	11.8

```
[19]: search_df.to_csv("./2650026_lab3_data/2650026_lab3_data_search.csv",
        index=False)
```

7 Q.

Delete an element from the array input from input.txt, and then store the new output in output.txt.

```
[20]: def AVL_Delete(value, root_node: AVL_Node | None) -> AVL_Node | None:
    if root_node is None or root_node.value is None:
        if PRINT_DEBUG:
            print(f"Value {value} not found in the AVL tree.")
        return root_node

    if value < root_node.value:
        root_node.lchild = AVL_Delete(value, root_node.lchild)
    elif value > root_node.value:
        root_node.rchild = AVL_Delete(value, root_node.rchild)
    else:
        # root_node IS the node to delete
        if root_node.lchild is None or root_node.rchild is None:
            child = root_node.lchild if root_node.lchild is not None else ↵
            ↵root_node.rchild
            if child is not None:
                # Splice child past root_node. root_node.parent may still be a
                # real ancestor (if root_node is internal) or None (if root_node
                # is the true tree root) - both are correct values to copy here,
                # BEFORE root_node goes out of scope. Same "capture before
                # mutate" issue as OG_PARENT in the rotation functions: without
                # this line, child.parent would keep dangling-pointing at the
                # just-deleted root_node whenever root_node was the whole tree's
                # root (the one case with no enclosing setter to auto-fix it).
                child.parent = root_node.parent
            if PRINT_DEBUG:
                print(f"Deleted {root_node}, spliced in {child}")
            return child # 0 or 1 child: caller's setter (or bare reassignment
                # at the true root) takes it from here
        else:
            # 2 children: find in-order successor (leftmost node of right ↵
            ↵subtree)
            successor = root_node.rchild
            while successor.lchild is not None:
                successor = successor.lchild
            if PRINT_DEBUG:
                print(f"Replacing {root_node} with successor value {successor.
            ↵value}")
            root_node.value = successor.value
            root_node.rchild = AVL_Delete(successor.value, root_node.rchild)

    # --- Rebalance, checked at EVERY level as recursion unwinds ---
```

```

# Unlike insert, a single rotation here does NOT guarantee ancestors above
# stay balanced - deletion can shrink a subtree's height, so this check has
# to run at every returning frame, potentially cascading multiple rotations
# all the way up to the root (O(log n) rotations in the worst case, vs. at
# most 1 for insert).
root_node.update_height_and_balance()

if root_node.balance_factor == 2:
    # child's bf uses >= 0 (not > 0 like insert) because deletion can leave
    # the taller child PERFECTLY balanced (bf == 0), which still requires a
    # single rotation, not a double one - insert can never produce that
    # exact situation, which is why this threshold differs from insert's.
    if root_node.lchild.balance_factor >= 0: # type: ignore
        root_node = AVL_right_rotate(root_node, root_node)          # LL
    else:
        root_node = AVL_double_rotate_right(root_node, root_node)   # LR
elif root_node.balance_factor == -2:
    if root_node.rchild.balance_factor <= 0: # type: ignore
        root_node = AVL_left_rotate(root_node, root_node)          # RR
    else:
        root_node = AVL_double_rotate_left(root_node, root_node)    # RL

return root_node

```

```
[21]: root0 = AVL_Delete(77, root0)
```

```
[22]: traversal_file = open("./2650026_lab3_data/output1.txt", 'w')
      AVL_inorder_traversal(root0, out_file = traversal_file)
      traversal_file.close()
```

```

25, Node(value=25, height=0, balance_factor=0)
50, Node(value=50, height=1, balance_factor=1)
55, Node(value=55, height=2, balance_factor=0)
67, Node(value=67, height=1, balance_factor=-1)
69, Node(value=69, height=0, balance_factor=0)
70, Node(value=70, height=4, balance_factor=-1)
76, Node(value=76, height=0, balance_factor=0)
90, Node(value=90, height=2, balance_factor=-1)
108, Node(value=108, height=0, balance_factor=0)
210, Node(value=210, height=1, balance_factor=0)
348, Node(value=348, height=0, balance_factor=0)
420, Node(value=420, height=3, balance_factor=0)
720, Node(value=720, height=0, balance_factor=0)
768, Node(value=768, height=1, balance_factor=1)
911, Node(value=911, height=2, balance_factor=1)
1080, Node(value=1080, height=0, balance_factor=0)

```